

NYC Yellow Taxi

Ingestion, ETL & Analytics on HDFS + Spark

COURSE & CLASS

Course — Big Data Analytics

Class — BSCS-13A

Course Group — Group 3 (BDA, BSCS-13A)

Project Group # — 17

Institution — NUST SEECS

GROUP 17 · CONTRIBUTIONS

Malik Muhammad Hassaan — CMS 460233 ·

Ingestion · Validation · Profiling · Cleaning strategy

Muhammad Ahmed — CMS 456679 · Spark ETL ·

Dimensional model · Analytics · Optimizations

Dataset Overview, Rationale & Business Value

DATASET FACTS

Source — NYC TLC Yellow Taxi (Jan 2024)

File — yellow_tripdata_2024-01.parquet

Volume — 2,964,624 trips · 19 columns

Format — Parquet — binary, columnar

WHY THIS DATASET

- Volume > 500K rows justifies HDFS + Spark
- Each row is a trip — natural fact table
- Clear dimensions: time, location, payment, vendor
- Real DQ issues — perfect ETL teaching dataset

WHY ANALYTICS ON THIS DATA MATTERS

- Reveals when and where demand spikes
- Quantifies how riders pay and tip
- Identifies highest-revenue origin-destination pairs
- Surfaces driver / vendor performance differences

BUSINESS BENEFITS DELIVERED

- Surge / dispatch windows → higher revenue
- Vehicle repositioning to hot OD pairs
- Card-payment investment justified by share data
- Tip insights → driver coaching & retention

Data Quality Issues & Cleaning Methods

Identified in profile_data.py · Resolved in etl.py · 0 nulls in keys after ETL.

ISSUE → ACTION

- **Negative fares** — median per distance bin (etl.py 84–93)
- **Zero passenger count** — round + clip min 1 (etl.py 96–99)
- **Non-integer passenger** — round to integer (etl.py 96–99)
- **Duplicate rows** — dropDuplicates (etl.py 71)
- **Distance outliers** — cap at 99th percentile (etl.py 110–114)

ISSUE → ACTION

- **Tip > fare** — cap tip at fare (etl.py 102–107)
- **Missing numerics** — median imputation (etl.py 117–125)
- **Missing categorical** — fallback to 0 / mode (etl.py 128–131)
- **Datetime inconsistencies** — to_timestamp + duration (etl.py 74–81)
- **Future timestamps** — median by hour/weekday (profile 86–88)

End-to-End Pipeline Architecture

Ingest

ingest.py

Download · validate
≥ 500K rows · upload
to HDFS

Profile

profile_data.py

Schema · missing ·
stats · DQ issues

ETL

etl.py (Spark)

Clean · derive · star
schema · partition

Analytics

analytics.py

5 Spark SQL queries ·
4 charts

Report

build_final_report

Bundle CSVs + charts
→ PDF

Storage: HDFS /warehouse/raw/... → /warehouse/processed/...

Compute: Apache Spark 3.5.8 · PySpark · Spark SQL · HDFS NameNode + DataNode local cluster

Validation: etl_summary.json · raw row count = fact_trips row count = 2,964,624 · zero nulls in keys

Dimensional Model — Star Schema

dim_datetime

749 rows · etl.py 164–169

dim_payment

5 rows · etl.py 171–184

fact_trips

2,964,624 rows

distance · fare · tip · revenue · IDs

dim_trip_category

3 rows · etl.py 186–195

Joined on natural keys: datetime_key · payment_type · distance_bucket · small dims → broadcast joins.

Five Business Questions (analytics.py)

Q1 — Trip volume & average fare by hour and day of week (lines 43–65)

Q2 — Correlation between trip distance and fare amount (lines 66–77)

Q3 — Payment-type preference by time and pickup location (lines 78–96)

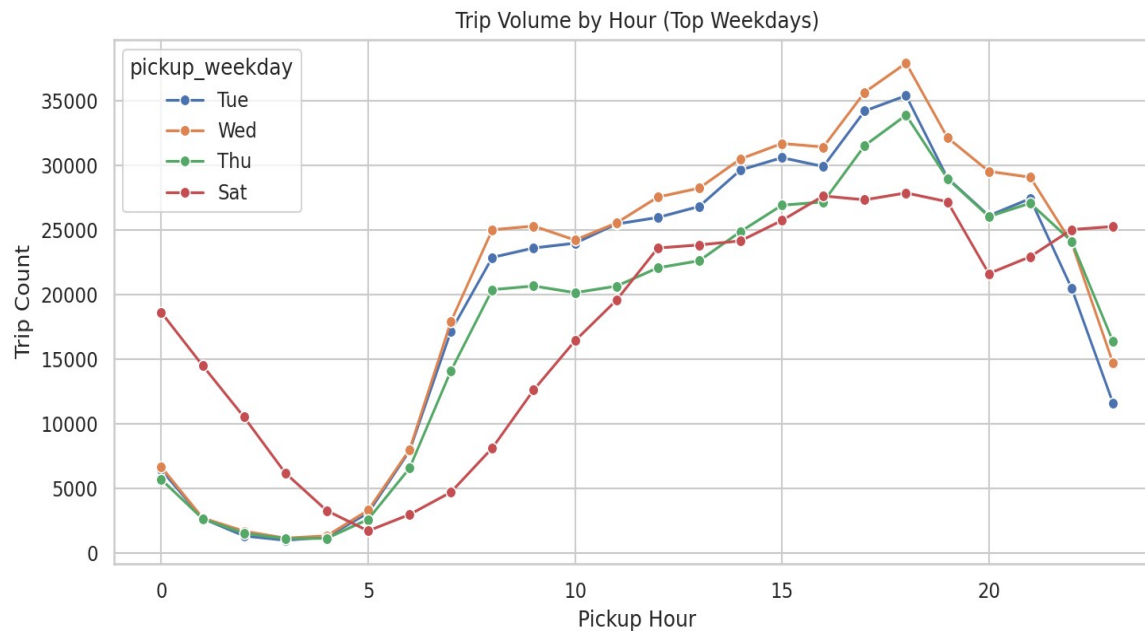
Q4 — Average tip by payment type, distance, and vendor (lines 97–111)

Q5 — Top pickup → dropoff pairs by volume and revenue (lines 112–126)

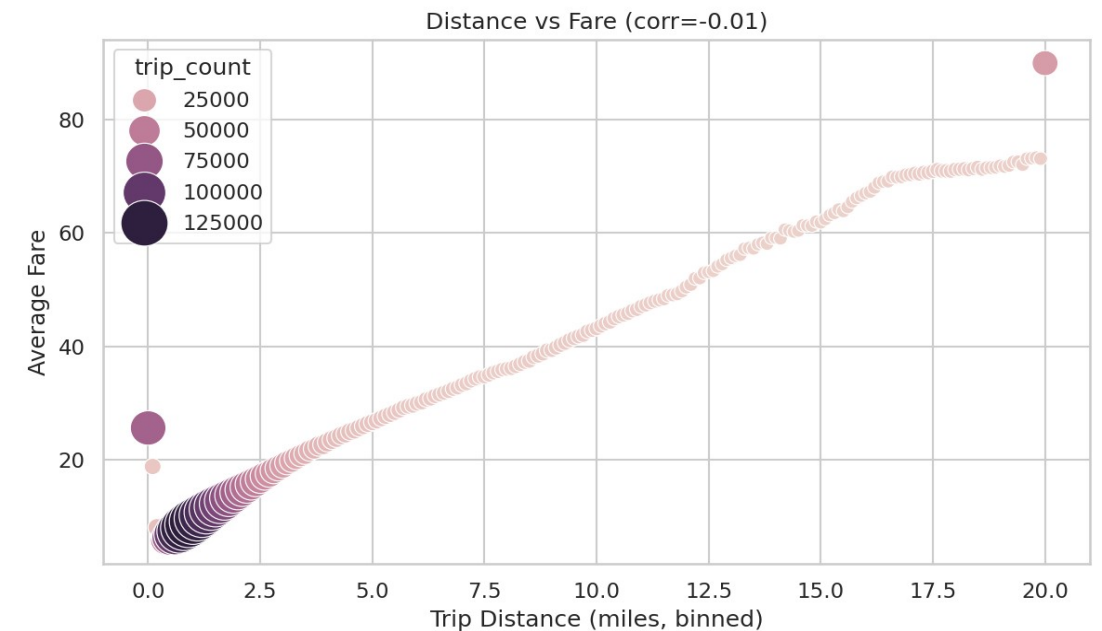
Source of questions: Dataset_Justification.pdf · Implemented as Spark SQL views over fact_trips and dims.

FINDINGS

Q1 — Hourly Demand • Q2 — Distance vs Fare



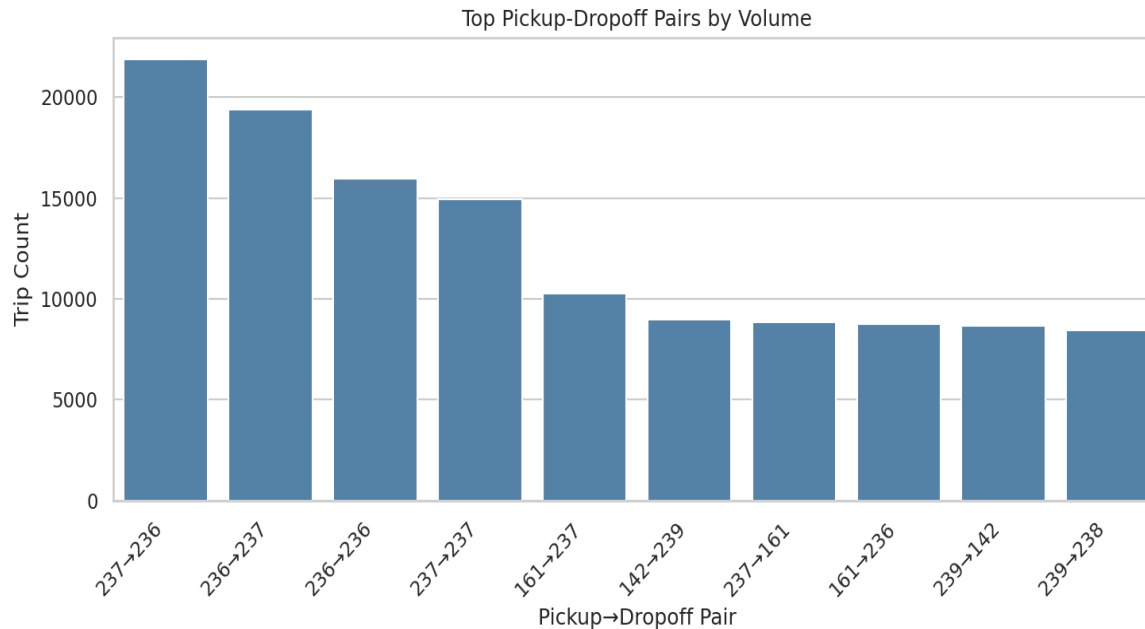
- Demand peaks 5–7 PM on weekdays.
- Fri/Sat sustain volume late at night.
- Lowest volume in the 3–5 AM window.



- Strong positive distance ↔ fare relationship.
- Within-bin correlation low → time/traffic noise.
- Distance is the dominant fare driver.

FINDINGS

Q3 Payment • Q4 Tipping • Q5 Top OD Pairs



KEY NUMBERS FROM CSVs

- **Q3 Payment** — Credit card 80–84% in zones 161/236/237
- **Q4 Tip CC long** — \$11.58 · 21.2% of fare
- **Q4 Tip CC short** — \$2.71 · 27.9% of fare
- **Q4 Cash / dispute** — $\approx$ \$0 (no-charge \$0.11)
- **Q5 Top volume** — 237 → 236 · 21,883 trips
- **Q5 Top revenue** — 132 → 132 · \$246,050 (JFK)

- Credit card dominates payment in core Manhattan during peak hours.
- Tip \$ highest on long trips; tip % highest on short trips.
- A handful of OD pairs (Manhattan core + JFK) drive most revenue.

Pipeline Optimizations

- **Broadcast joins** — small dims wrapped in `F.broadcast(...)` (etl.py 198–200)
- **Cache reused DF** — `transformed.cache()` before deriving fact + dims (etl.py 161)
- **Partitioned Parquet** — `fact_trips` & `dim_datetime` partitioned by year, month (etl.py 370, 373)
- **Approx percentile** — `percentile_approx` for distance cap & medians (etl.py 110, 117–122)
- **Dynamic partition overwrite** — `partitionOverwriteMode=dynamic` (etl.py 60)
- **Tuned shuffle partitions** — `spark.sql.shuffle.partitions = 16` (etl.py 59)

Business Insights & Conclusion

- **Demand is time-bound** — Peak 5–7 PM, dawn lull → dynamic dispatch / surge windows.
- **Distance dominates fare** — Strong positive trend → predictable distance-based pricing.
- **NYC taxis are cashless** — Credit card 80%+ in busiest zones → invest in card UX.
- **Tipping is payment-driven** — Card riders tip materially; cash / disputes ≈ 0 .
- **Revenue concentrated** — Manhattan core (161/236/237) + JFK (132) drive most revenue.
- **Data quality discipline** — All 2.96M rows preserved · 0 nulls in keys (etl_summary.json).

Dataset & GitHub Repository

DATASET

NYC TLC Yellow Taxi Trip Records (Jan 2024)

<https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page>

File: yellow_tripdata_2024-01.parquet

GITHUB REPOSITORY

NYC-Taxi-Ingestion-Pipeline

<https://github.com/hassaanawan48/NYC-Taxi-Ingestion-Pipeline>

Includes ingest.py · profile_data.py · etl.py · analytics.py · build_final_report.py

Thank you · Questions are welcome.